Matrices & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reciprocal, quotient, and Pythaogrean identities**

Given the matrices $A = [4, -2; 1, 3]$ and $B = [-1, 5; 2, 0]$, what is the matrix sum $A + B$?

To add two matrices of the same dimension, add their corresponding entries: - Row 1, Column 1: $4 + (-1) = 3$ - Row 1, Column 2: $-2 + 5 = 3$ - Row 2, Column 1: $1 + 2 = 3$ - Row 2, Column 2: $3 + 0 = 3$ Therefore, $A + B = [3, 3; 3, 3]$. Common misconceptions: - Subtracting entries instead of adding yields $[5, -7; -1, 3]$. - Multiplying corresponding entries entrywise yields $[-4, -10; 2, 0]$.

2 **[Foundational]** • **Target: Cofunction identities**

Given the matrices $A = (3, -2; 1, 4)$ and $B = (-5, 2; 0, -3)$, what is the matrix $2A - B$?

To find $2A - B$, first compute the scalar multiple $2A$ by multiplying every entry of A by 2: $2A = 2(3, -2; 1, 4) = (2(3), 2(-2); 2(1), 2(4)) = (6, -4; 2, 8)$ Next, subtract matrix B from $2A$ entry by entry: $2A - B = (6 - (-5), -4 - 2; 2 - 0, 8 - (-3)) = (6 + 5, -6; 2, 8 + 3) = (11, -6; 2, 11)$ Common errors: - Adding the matrices instead of subtracting gives $2A + B = (1, -2; 2, 5)$. - Forgetting to multiply A by the s.

3 **[Foundational]** • **Target: Angle sum and difference identities**

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ Substitute the given values into the equation: $6^2 = x(x + 5)$ $36 = x^2 + 5x$ $x^2 + 5x - 36 = 0$ Factor the quadratic equation: $(x + 9)(x - 4) = 0$ This yields $x = -9$ or $x = 4$. Because x represents a geometric length, it must be positive, so $x = 4$. The length of hypotenuse AB is the sum of segments AD and DB.

4 **[Foundational]** • **Target: Verifying trig identities**

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

Let a_1 be the first term and r be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1 r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1 r + a_1 r^2 + a_1 r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for S_4: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for a_1.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Reciprocal, quotient, and Pythagorean identities

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

To find $\sin(\theta)$ for an angle in standard position whose terminal side passes through $(x, y) = (-5, 12)$: 1. Calculate the distance r from the origin to the point: $r = \sqrt{x^2 + y^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. By the coordinate definition of trigonometric functions: $\sin(\theta) = \frac{y}{r} = \frac{12}{13}$ Misconception checks: - $-\frac{5}{13}$ results from confusing sine with cosine, where $\cos(\theta) = \frac{x}{r}$. - $-\frac{12}{13}$ comes from mistakenly as.

6 [Foundational] • Target: Cofunction identities

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?

Let the right-angled triangle be denoted as ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, and its length is given as 5 units. We are looking for the length of the hypotenuse AB. The sine of an angle in a right-angled triangle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. Therefore, $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB}$. We have $\sin(30^\circ) = \frac{5}{AB}$. We know that.

7 [Foundational] • Target: Angle sum and difference identities

Given the matrices $A = (5, -1; 2, 4)$ and $B = (k, 2; -2, 5)$, for what value of the constant k is the matrix $C = 2A - B$ not invertible?

To determine the value of k for which $C = 2A - B$ is not invertible, proceed in three steps: 1. First, calculate the scalar multiple $2A$: $2A = 2(5, -1; 2, 4) = (10, -2; 4, 8)$ 2. Next, subtract matrix B from $2A$ to find C: $C = 2A - B = (10, -2; 4, 8) - (k, 2; -2, 5) = (10 - k, -2 - 2; 4 - (-2), 8 - 5) = (10 - k, -4; 6, 3)$ 3. A square matrix is not invertible if and only if its determinant is equal to zero.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying trig identities

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

The general term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n - 1)d$ Substitute $a_1 = 5$, $d = 3$, and $n = 6$: $a_6 = 5 + (6 - 1) \cdot 3 = 5 + 5 \cdot 3 = 5 + 15 = 20$ Misconception Analysis: - 23 comes from using $a_1 + n \cdot d = 5 + 6(3) = 23$, forgetting to subtract 1 from n. - 17 is a_5, resulting from an off-by-one counting error. - 18 comes from directly multiplying $n \cdot d = 6 \cdot 3 = 18$, neglecting the initial term.

© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

What are the coordinates of the center and the radius of the circle given by the equation $(x-4)^2+(y+1)^2=36$?

The standard form of the equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where (h, k) is the center and r is the radius. Matching the given equation $(x-4)^2+(y+1)^2=36$ to the standard form: 1. $x-h=x-4 \Rightarrow h=4$ 2. $y-k=y-(-1) \Rightarrow k=-1$ 3. $r^2=36 \Rightarrow r=\sqrt{36}=6$ Therefore, the center is $(4,-1)$ and the radius is 6. Common misconceptions: - Forgetting the negative signs in $(x-h)$ and $(y-k)$ leads to the incorrect center $(-4, 1)$. - Forg.

10 [Intermediate] • Target: Cofunction identities

A small bakery recorded the number of muffins sold each day for a week. The results are shown in the bar chart below. What is the total number of muffins sold during the week?

The bar chart shows the number of muffins sold for each day of the week. To find the total number of muffins sold, we sum the sales for each day: Monday (10) + Tuesday (5) + Wednesday (15) + Thursday (8) + Friday (7) + Saturday (5). Total = 10 + 5 + 15 + 8 + 7 + 5 = 50 muffins.

11 [Intermediate] • Target: Angle sum and difference identities

The polynomial $P(x)=2x^3+ax^2+bx-6$ has a factor of $(x-2)$. When $P(x)$ is divided by $(x+1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x-3)$?

We use the Factor Theorem and the Remainder Theorem: 1. Since $(x-2)$ is a factor of $P(x)$, we have $P(2)=0$: $P(2)=2(2)^3+a(2)^2+b(2)-6=16+4a+2b-6=4a+2b+10=0$ $2a+b=-5 \quad---$ (Equation 1) 2. Since the remainder when divided by $(x+1)$ is -12, we have $P(-1)=-12$: $P(-1)=2(-1)^3+a(-1)^2+b(-1)-6=-2+a-b-6=a-b-8=-12$ $a-b=-4 \quad---$ (Equation 2) 3. Add Equation 1 and Equation 2 to eliminate b.

12 [Intermediate] • Target: Verifying trig identities

An ellipse in the xy-plane is defined by the equation $9x^2+16y^2-54x+64y+c=0$ where c is a constant. If the length of the minor axis of the ellipse is 6, what is the value of c?

To find c, convert the given equation into standard form by completing the square for both x and y: 1. Group the x and y terms: $(9x^2-54x)+(16y^2+64y)=-c$ 2. Factor out the leading coefficients: $9(x^2-6x)+16(y^2+4y)=-c$ 3. Complete the square inside each set of parentheses: $9(x^2-6x+9)+16(y^2+4y+4)=-c+9(9)+16(4)$ $9(x-3)^2+16(y+2)^2=145-c$ 4. Divide by $145-c$ to write the equation in standard form.

© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

By the Factor Theorem, if $x = 2$ and $x = -3$ are zeroes of $P(x)$, then $P(2) = 0$ and $P(-3) = 0$. 1. Substitute $x = 2$: $P(2) = (2)^4 + a(2)^3 - 7(2)^2 + b(2) - 12 = 0$

$16 + 8a - 28 + 2b - 12 = 0 \Rightarrow 8a + 2b - 24 = 0 \Rightarrow 4a + b = 12$ (Equation 1) 2. Substitute $x = -3$:

$P(-3) = (-3)^4 + a(-3)^3 - 7(-3)^2 + b(-3) - 12 = 0$ $81 - 27a - 63 - 3b - 12 = 0 \Rightarrow -27a - 3b + 6 = 0 \Rightarrow 9a + b = 2$ (Equation 2) 3. Solve the system of linear equations: Subt.

14 [Intermediate] • Target: Cofunction identities

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

The sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$. If the sequence converges to a limit L, then as $n \to \infty$, we have $a_{n+1} \to L$ and $a_n \to L$. Substituting L into the recurrence relation, we get: $L = \frac{L^2 + 4}{2L}$ Multiplying both sides by $2L$ (assuming $Leq0$), we have: $2L^2 = L^2 + 4$ $L^2 = 4$ This gives $L = 2$ or $L = -2$. Let's check the first few terms of the sequence to determine the sign of the limit. $a_1 = 3$..

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of h and k.

The standard form of a horizontal hyperbola centered at (h, k) is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. The vertices of such a hyperbola are located at $(h \pm a, k)$. Given the vertices are $(5, 2)$ and $(-1, 2)$, we can equate these to $(h + a, k)$ and $(h - a, k)$ respectively. From the y-coordinates of the vertices, we see that $k = 2$. From the x-coordinates, we have two equations: 1) $h + a = 5$ 2) $h - a = -1$ Adding these two equations together, we get $(h + a) + (h - a) = 5 + (-1)$, which simplifies to.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 6.01 Radicals & Rational Exponents

For all real values of x such that $xeq-1$ and $xeq2$, the equation $\frac{ax+7}{x+1}-\frac{b}{x-2}=\frac{2x^2-3x-20}{x^2-x-2}$ is true, where a and b are constants. What is the value of $a+b$?

To find the values of a and b, combine the fractions on the left-hand side over their common denominator, $(x+1)(x-2)=x^2-x-2$: $\frac{(ax+7)(x-2)-b(x+1)}{(x+1)(x-2)}=\frac{2x^2-3x-20}{x^2-x-2}$ Since the denominators are equal for all $xeq-1,2$, their numerators must be identical for all such values of x:
$(ax+7)(x-2)-b(x+1)=2x^2-3x-20$ Expand the left side:
$ax^2-2ax+7x-14-bx-b=2x^2-3x-20$ $ax^2+(7-2a-b)x-(14+b)=2x^2-3x-20$ Equ.

17 [Challenging] • Target: Reciprocal, quotient, and Pythaogrean identities • Cross-Lesson: 7.01 Sequences & Series

A manufacturing process produces items, and the number of defects per batch of 100 items follows a Poisson distribution with a mean of λ. The cost of inspecting a batch is C_i dollars, and the cost of repairing a defective item is C_r dollars. If a batch has k defects, the total cost for that batch is $C(k)=C_i+k\cdot C_r$. If $\lambda=3$, $C_i=50$, and $C_r=10$, find the expected total cost per batch, given that the expected number of defects in a batch is $E[K]=\lambda$.

Let K be the random variable representing the number of defects in a batch of 100 items. K follows a Poisson distribution with mean λ. The cost of inspecting a batch is C_i, and the cost of repairing a defective item is C_r. The total cost for a batch with k defects is $C(k)=C_i+k\cdot C_r$. We are given $\lambda=3$, $C_i=50$, and $C_r=10$. We need to find the expected total cost per batch, $E[C(K)]$. Using the linearity of expectation: $E[C(K)]=E[C_i+K\cdot C_r]$.

18 [Challenging] • Target: Cofunction identities • Cross-Lesson: 8.01 Conic Sections

Consider the equation $\sqrt{2x+k}=x-1$, where k is a constant. If $x=5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

Step 1: Find the value of the parameter k by substituting the valid solution $x=5$ into the equation:
$\sqrt{2(5)+k}=5-1\Rightarrow\sqrt{10+k}=4$ Squaring both sides gives $10+k=16$, so $k=6$. Step 2: Substitute $k=6$ back into the equation and solve by squaring both sides: $\sqrt{2x+6}=x-1$
$2x+6=(x-1)^2=x^2-2x+1$ $x^2-4x-5=0$ Factoring the quadratic equation gives:
$(x-5)(x+1)=0\Rightarrow x=5$ or $x=-1$ Step 3: Check for extraneous sol.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 9.01 Trigonometric Functions

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

To find the z-score corresponding to a given data value, use the standard normal distribution formula: $z = \frac{x - \mu}{\sigma}$ where x is the individual score, μ is the population mean, and σ is the standard deviation. Substitute the given values into the formula: $z = \frac{58 - 74}{8} = \frac{-16}{8} = -2$ A z-score of -2 indicates that the score of 58 is located 2 standard deviations below the mean. Common Misconceptions: - 2: Results from subtracting the score from the mean.

20 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 1.01 Functions & Their Graphs

Let $f(x) = \sqrt{x^2 - 6x + 9}$ and $g(x) = x - 3$. For which values of x is $f(x) = g(x)$?

First, simplify the expression for $f(x)$. We recognize that the radicand, $x^2 - 6x + 9$, is a perfect square trinomial, which can be factored as $(x - 3)^2$. Therefore, $f(x) = \sqrt{(x - 3)^2}$. The definition of the square root of a square is $\sqrt{a^2} = |a|$. Applying this, we get $f(x) = |x - 3|$. Now, we need to solve the equation $f(x) = g(x)$, which becomes $|x - 3| = x - 3$. The absolute value of an expression is equal to the expression itself if and only if the expression is non-negative. Thus, $|x - 3| = x - 3$ if.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

22 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(5)$.

a. 15 ✓

b. 17

c. 10

d. 19

Substitute $x = 5$: $f(5) = 2(5) + 5 = 15$.

24 [Foundational] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

25 [Foundational] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

26 [Foundational] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(5)$.

a. 21 ✓

b. 24

c. 15

d. 25

Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

27 [Foundational] • Target: Angle sum and difference identities

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 45^\circ) = 88^\circ$.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Angle sum and difference identities

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

30 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

A data set recorded for reciprocal, quotient, and pythaogrean identities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

31 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(4)$.

a. 13 ✓

b. 15

c. 8

d. 17

Substitute $x = 4$: $f(4) = 2(4) + 5 = 13$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Reciprocal, quotient, and Pythaogrean identities

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

33 [Intermediate] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

34 [Intermediate] • Target: Cofunction identities

A data set recorded for cofunction identities consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

35 [Intermediate] • Target: Cofunction identities

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(4)$.

a. 18 ✓

b. 21

c. 12

d. 22

Substitute $x = 4$: $f(4) = 3(4) + 6 = 18$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Cofunction identities

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

37 [Intermediate] • Target: Angle sum and difference identities

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 8.01 Conic Sections

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12 ✓

d. 17

Substituting $x = 1$ into $y = 5x + 7$ gives $y = 5(1) + 7 = 12$.

39 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 9.01 Trigonometric Functions

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17 ✓

d. 22

Substituting $x = 2$ into $y = 5x + 7$ gives $y = 5(2) + 7 = 17$.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 1.01 Functions & Their Graphs

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74° ✓

b. 84°

c. 66°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 59^\circ) = 74^\circ$.

41 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 11.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

42 [Challenging] • Target: Angle sum and difference identities • Cross-Lesson: 2.01 Quadratics & Complex Numbers

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 22

b. 29

c. 27 ✓

d. 32

Substituting $x = 4$ into $y = 5x + 7$ gives $y = 5(4) + 7 = 27$.

43 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for verifying trig identities consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 5x + 8$ in verifying trig identities, evaluate $f(3)$.

a. 23 ✓

b. 28

c. 15

d. 27

Substitute $x = 3$: $f(3) = 5(3) + 8 = 23$.

45 [Challenging] • Target: Verifying trig identities • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved